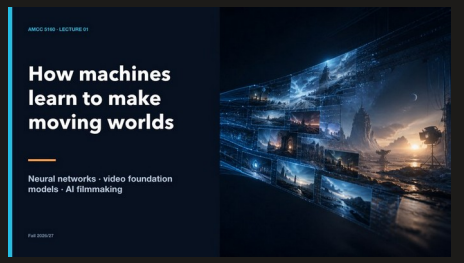
AMCC 5160 · LECTURE 11 · 10 NOVEMBER 2026

# The Semester *in One Evening*

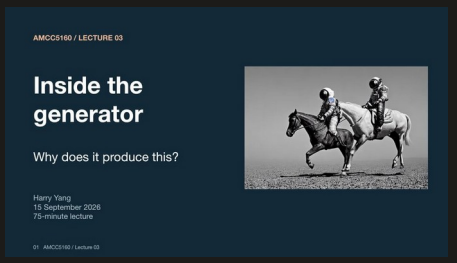
A review of AMCC 5160: ten lectures, four threads, one habit

Review session · about one hour · Fall 2026

Harry Yang · Division of Arts and Machine Creativity, HKUST · MIL13002




Lecture 01




Lecture 03




Lecture 05




Lecture 07




Lecture 08




Lecture 09

# Tonight: one hour, four threads

| 5 | 12 | 10 | 13 | 10 | 10 |
|---|---|---|---|---|---|
| min | min | min | min | min | min |
| **Opening** | **Thread 1** | **Thread 2** | **Thread 3** | **Thread 4** | **Wrap-up** |
| The semester on one line | How a generator works | Control: editing and making it yours | Time: minutes, voices, worlds | Judgment: looking, measuring, signing | Game, your four votes, final presentation |

After the break: the last weekly presentations

# Ten Tuesdays on one line

| | | | | |
|---|---|---|---|---|
| 01 · Sep 1 | 02 · Sep 8 | 03 · Sep 15 | 04 · Sep 22 | 05 · Sep 29 |
| **Neural networks → AI filmmaking** | **Generative editing** | **Inside the generator** | **Video generation + art filmmaking** | **Making the model yours** |
| *A generated video is a learned prediction about change* | *Write the edit contract: change, keep, adapt* | *The model predicts; the sampler takes a step* | *Two kinds of time; intention comes first* | *Whose model is it?* |
| 06 · Oct 6 | 07 · Oct 13 | 08 · Oct 20 | 09 · Oct 27 | 10 · Nov 3 |
| **Quantization (TA talk)** | **Music, minutes, machines** | **AI video that talks** | **Worlds, 3D, robots** | **Critique + provenance** |
| *Store less: fewer bits per number* | *The unit is now the scene; keep your sources* | *One model, one clock; who is the performer?* | *Add an action and video becomes a place* | *Is it AI? good? fair? allowed?* |

Weeks 12–13: final presentations · Nov 17 and Nov 24

# Four questions kept coming back

## 1

### How does it work?

Lectures 01 · 03 · 04

Data, latents, attention, noise, sampling

## 2

### How do I control it?

Lectures 02 · 05

Edit contracts, controls, LoRA, DreamBooth

## 3

### What happens over time?

Lectures 07 · 08 · 09

Minutes, sound, live characters, worlds

## 4

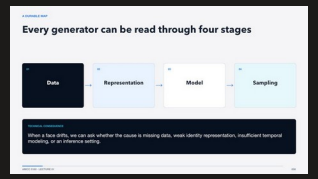
### Is it good, and is it mine?

Lectures 05 · 08 · 10

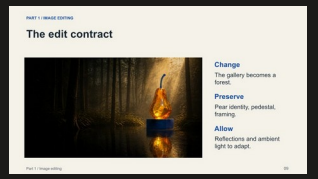
Critique, metrics, consent, credits

REVISIT

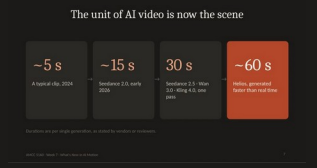
Click to open



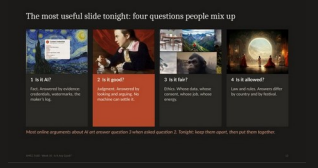

L01 · 8




L02 · 9




L07 · 7



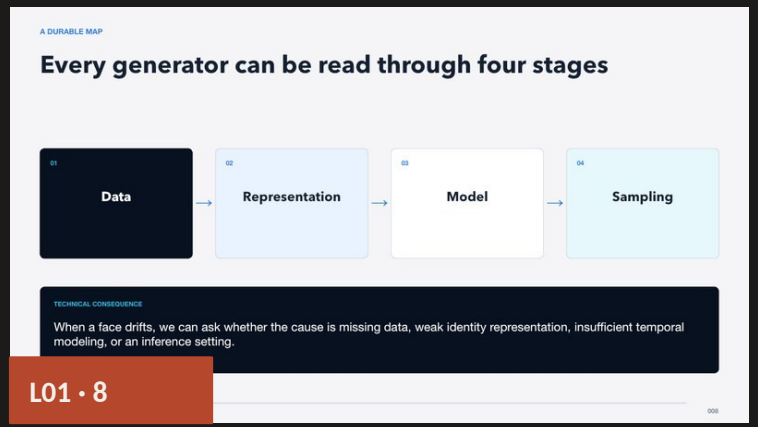

L10 · 13

1

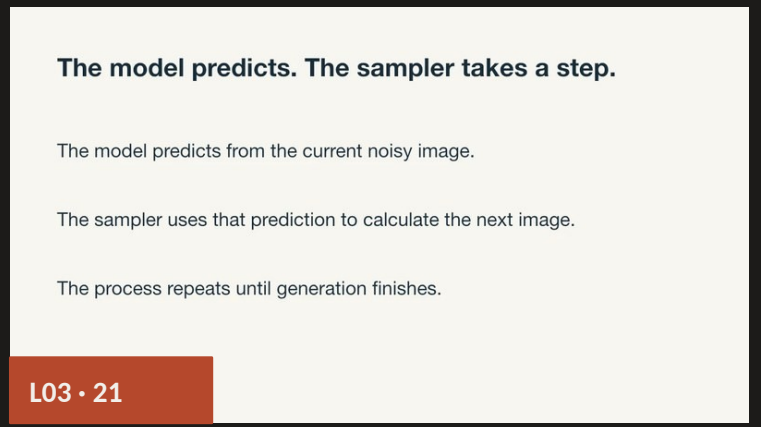
THREAD 1

# How a generator works

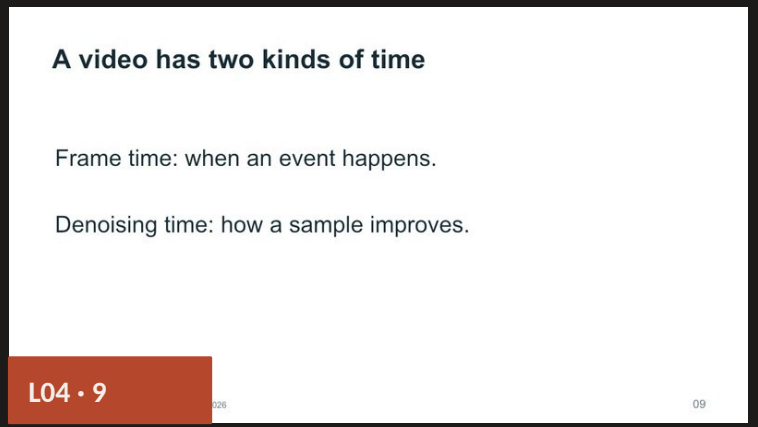
*Lectures 01, 03 and 04*




L01 · 8




L03 · 21




L04 · 9

# Every generator in four stages

## Data

Examples, captions, filters

*Lecture 04: filtering trades coverage for quality*

→

## Representation

VAE latents, patch tokens

*Lecture 01: compress first, then create*

→

## Model

Transformer with attention

*Lecture 03: predicts noise or velocity*

→

## Sampling

Seed, guidance, steps, sampler

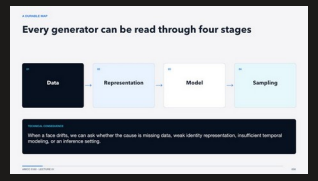
*Lecture 03: the sampler takes the step*

**DIAGNOSE**

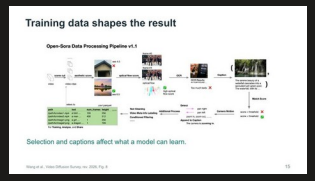
A face drifts. Which stage would you test first, and how?

**REVISIT**

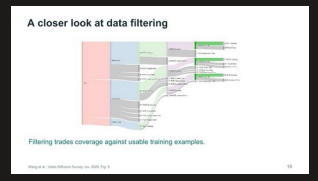
Click to open




L01 · 8




L04 · 15




L04 · 16

# Compress first, then create

**240**

frames in 10 s at 24 fps

Lecture 01

**16× · 4×**

spatial · temporal compression in H3

Lecture 01

**6× → ~36×**

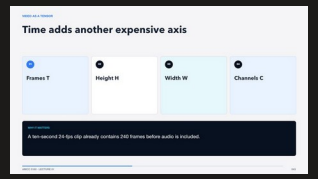
longer clip → attention comparisons

Lecture 07

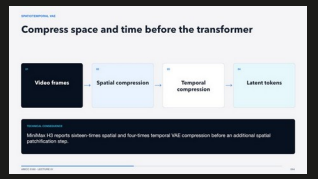
*Compression is a design requirement, not an optimisation. It decides which details can survive.*

REVISIT

Click to open




L01 · 43




L01 · 44




L07 · 9

# Learn to remove noise, then learn how to move

### Diffusion: add noise

$$x_t = a \cdot x_0 + b \cdot \varepsilon$$

$0.6 \times 0.8 + 0.8 \times (-0.2) = 0.32$

The network learns to predict ε, at every noise level

### Flow matching: learn to move

$$x_t = (1 - t) \cdot \varepsilon + t \cdot x$$

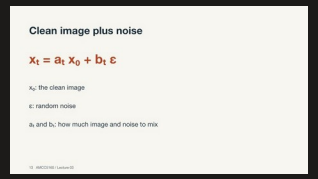
velocity = $x - \varepsilon$

At generation time there is no target; follow the learned field

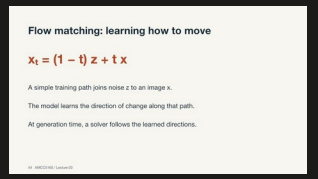
**The model predicts. The sampler takes the step.**

REVISIT

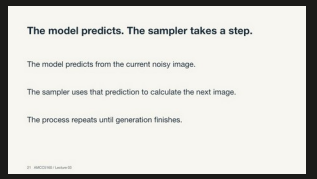
Click to open




L03 · 13

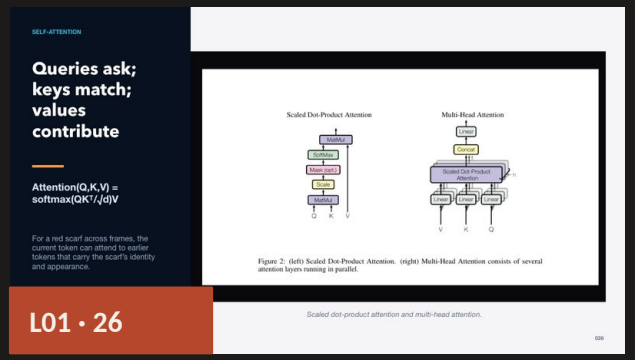



L03 · 44

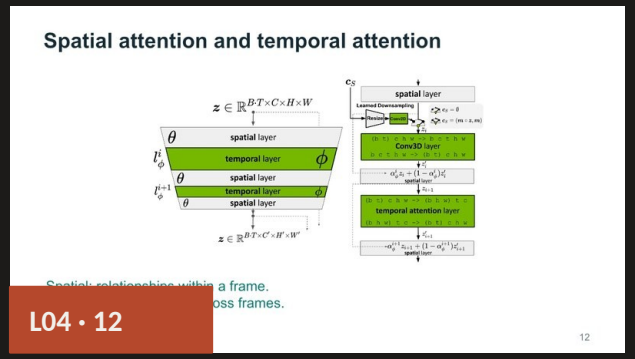



L03 · 21

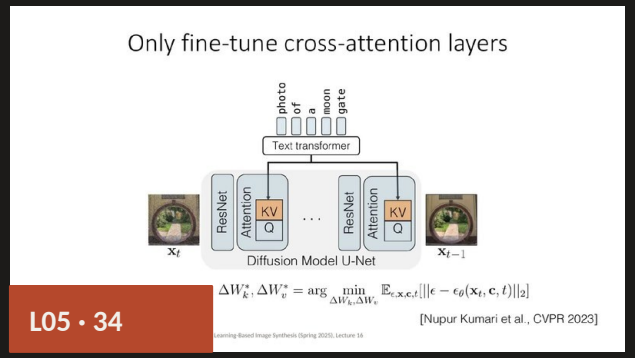
# Attention: every token chooses what matters



L01 · 26



L04 · 12



L05 · 34

1 **Queries ask, keys match, values contribute**

Softmax weights sum to one. Many heads learn different relations. (L01, L03)

2 **Cross-attention is where words meet pixels**

Custom Diffusion fine-tunes only its K and V, and still learns a concept. (L05)

3 **Spatial within a frame, temporal across frames**

Connecting features is not the same as understanding the scene. (L04)

# Three knobs, three different jobs

## Seed

An address, not an instruction

Save model, prompt, references and settings, not just the number.

L03 · 48–49

## Guidance

$\varepsilon = u + s \cdot (c - u)$

$u = 2$, $c = 3$, $s = 4 \rightarrow 6$. It extrapolates: stronger subject, burnt colours, less variety.

L03 · 52–55

## Steps

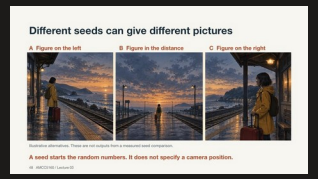
Accuracy of the path, not of the map

More steps ≠ more correct. A distilled 1–4 step model is a different trained system.

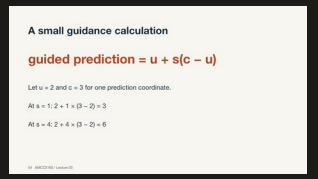
L03 · 60–64

REVISIT

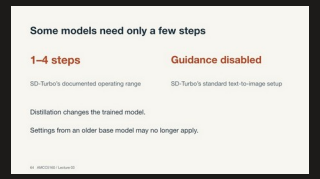
Click to open




L03 · 48




L03 · 54




L03 · 64

# Diagnose it: which knob, or no knob at all?

**A** Same prompt, a completely different composition every run

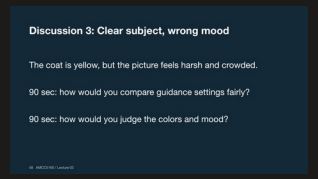
**B** Subject unmistakable, but colours burnt and every sample alike

**C** Looks fine, but 40 seconds per result, and you need it live

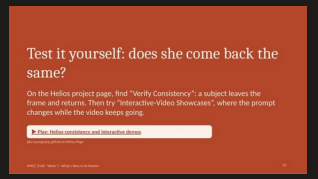
**D** A face changes after the character turns away and back

REVISIT

Click to open




L03 · 58




L07 · 22

# Diagnosis: three knobs and one memory problem

A

## Seed

Fix it to compare; record the whole workflow; use a reference for layout.

B

## Guidance too high

Lower the scale; compare across the same seeds; one variable at a time.

C

## Cost

Distilled model, fewer steps, quantization (Week 6). Measure seconds.

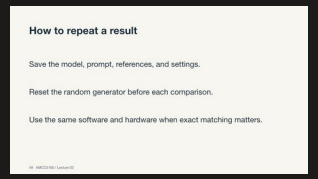
D

## Not a knob: memory

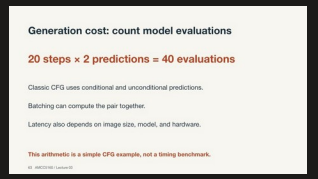
Representation and context over time. See Thread 3.

REVISIT

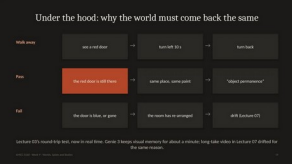
Click to open




L03 · 49




L03 · 63

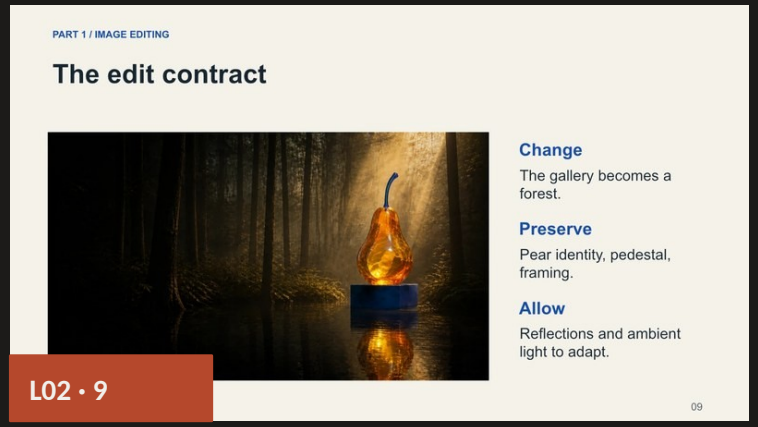



L09 · 19

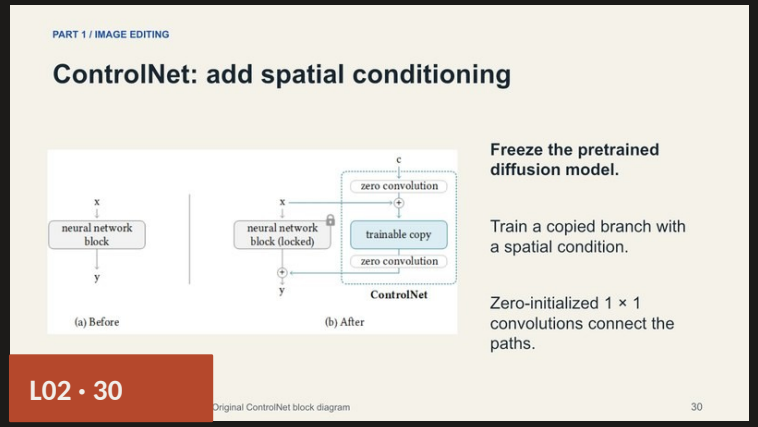
THREAD 2

# 2 Control: editing and making it yours

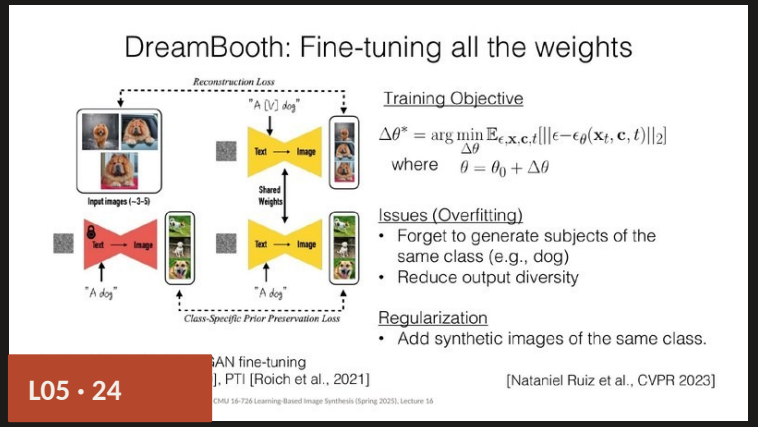
*Lectures 02 and 05*




L02 · 9




L02 · 30




L05 · 24

# The edit contract: change, keep, adapt

## Change

What the brief asks for

The environment: gallery → forest

## Keep

Invariants: semantic or exact?

The pear's identity, pedestal, framing

## Adapt

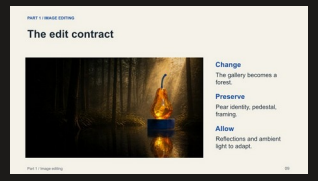
Consequences that must follow

Light, shadows, reflections

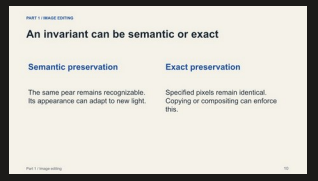
*“Keep it the same” can mean recognizable or pixel-identical. Only copying or compositing can promise pixel-identical.*

**REVISIT**

Click to open




L02 · 9



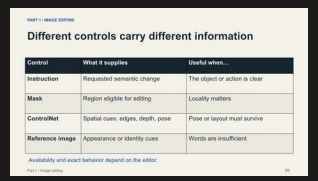

L02 · 10

# Choose a control by the information that is missing

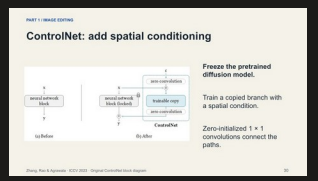
| Control | It supplies… | Watch out for… |
|---|---|---|
| **Text prompt** | What should change | A condition, not a hard constraint |
| **Mask** | Where change is allowed | Seams and reflections are not solved by the mask |
| **ControlNet: edges · depth · pose** | Shape and structure to follow | Says nothing about material or identity |
| **Reference image · IP-Adapter** | Appearance words can't pin down | Instant, but less faithful than training |
| **LoRA · DreamBooth** | A concept that must recur | Did it learn the subject, or the setting? |

REVISIT

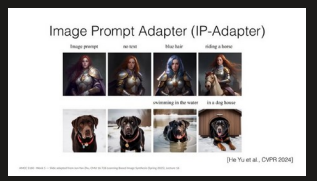
Click to open




L02 · 29




L02 · 30



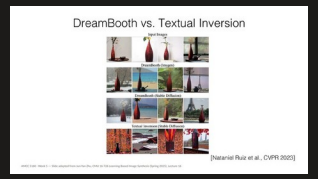

L05 · 64

# Making the model yours: what actually gets trained?

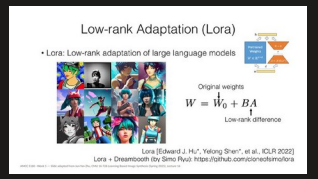
| Method | What is trained | Best for · cost |
|---|---|---|
| **Textual Inversion** | One new word embedding; model frozen | Styles · tiny file, weak identity |
| **DreamBooth** | All weights, with class-prior regularization | Identity · heavy, can overfit |
| **Custom Diffusion** | Only K and V in cross-attention | Combining concepts · small update |
| **LoRA** | A low-rank overlay: 16.8 M → 131 k per layer | The artist's default · swap overlays |
| **IP-Adapter** | Nothing per subject: an image encoder | Instant references · less faithful |

REVISIT

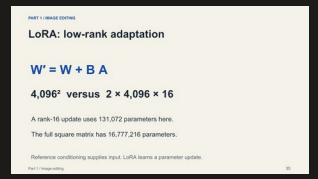
Click to open




L05 · 27



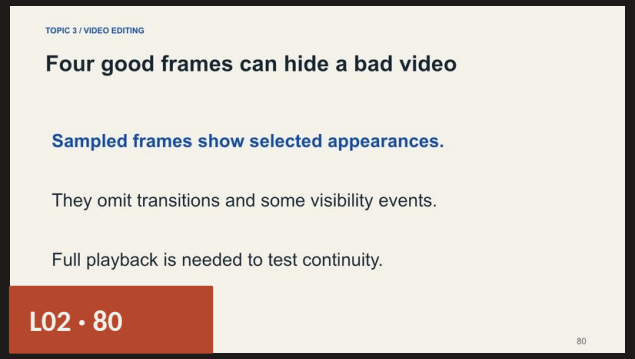

L05 · 61

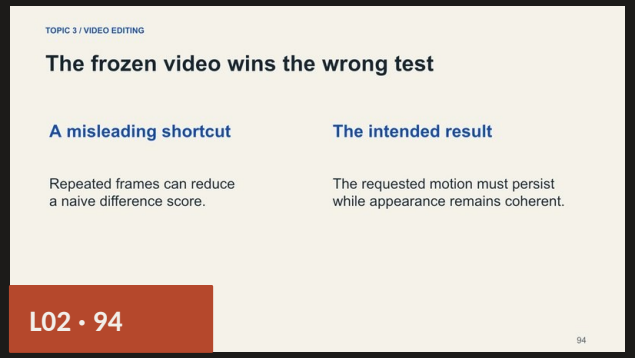



L02 · 35

# A video edit must survive the next frame




1 **Consistency is not stillness**

Change should be explained by the scene: pose, light, occlusion. (L02 · 76)

2 **Four good frames can hide a bad video**

Contact sheets locate problems; full playback finds them. (L02 · 80)

3 **The frozen video wins the wrong test**

Raw frame difference rewards a clip that never moves. (L02 · 93–94)

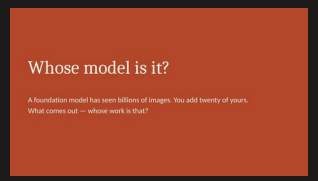
# “Whose model is it?”

The company that trained it, you who adapted it, or the people whose images are inside it?

Show of hands, 29 September. Has your hand moved?

**REVISIT**

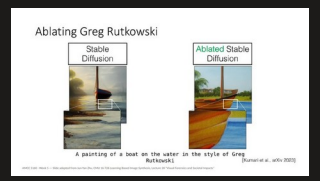
Click to open




L05 · 3




L05 · 30

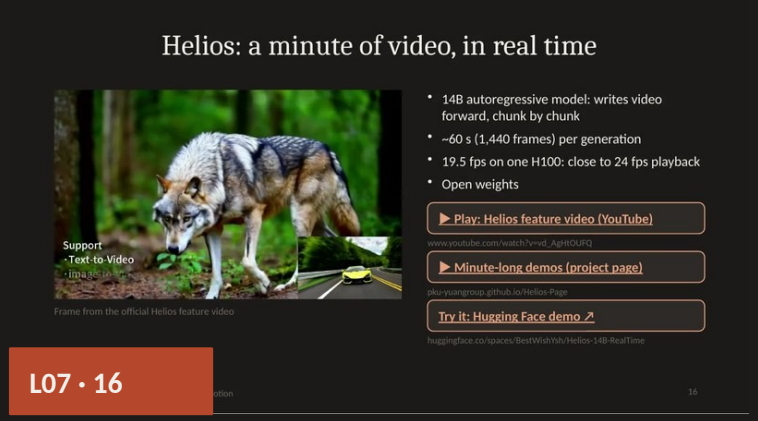



L05 · 54

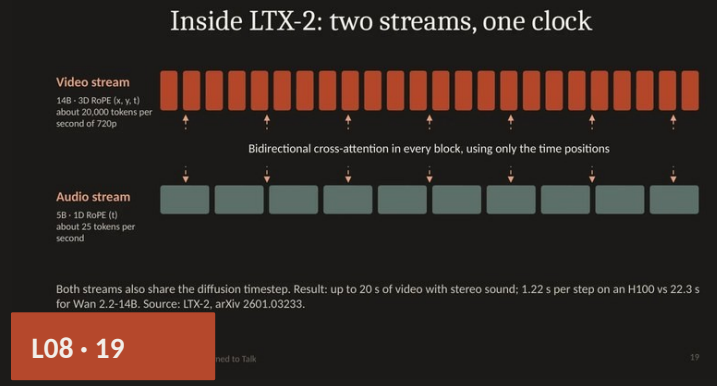
THREAD 3

# 3 Time: minutes, voices, worlds

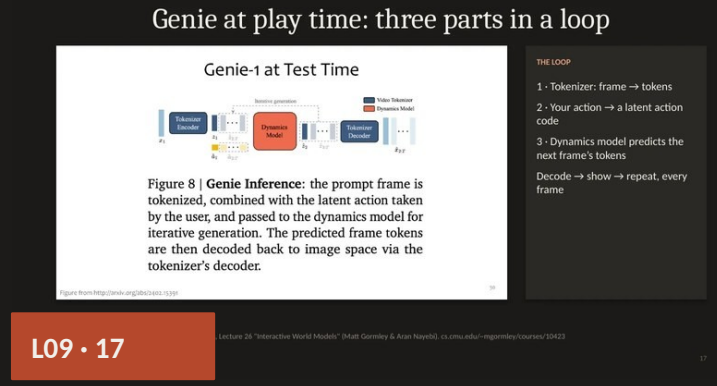
*Lectures 07, 08 and 09*




L07 · 16




L08 · 19




L09 · 17

# From the shot to the scene

**~5 s**

**2024**

A typical clip

**~15 s**

**Early 2026**

Seedance 2.0

**30 s**

**Summer 2026**

Seedance 2.5 · Wan 3.0 · Kling 4.0

**~1 min**

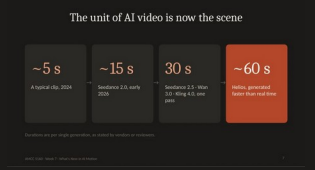
**Real time**

Helios: ~19.5 fps on one H100

*All at once: consistent, expensive. Forward in time: streams, but must fight drift.*

**REVISIT**

Click to open




L07 · 7




L07 · 10




L07 · 16

# Drift: one problem in three costumes

## Long video

Lecture 07

Trained on perfect frames, generating from its own. A jacket's shade drifts by second 60.

**Self Forcing · Helios trains on drifted video**

## World models

Lecture 09

Turn away from the red door, turn back: it is blue. Memory is the context window.

**Genie 3: about a minute of memory**

## Robots

Lecture 09

A small error leads somewhere the expert never went. Errors compound.

**Practice (RL) on top of copying**

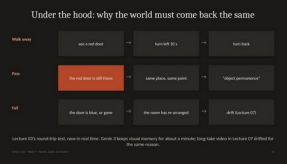
*Train on clean data, then live on your own mistakes. The fix: train on your own outputs, remember, practise.*

REVISIT

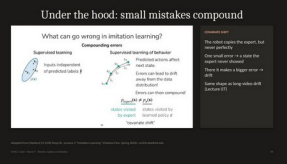
Click to open




L07 · 21




L09 · 19




L09 · 48

# The real-time toolkit, gathered from four lectures

## Change less

**Adapters · LoRA**

A thin overlay instead of the whole model

L05

## Step less

**Distillation**

Helios ~50 → 3 steps · TalkingMachines 24 → 2 evaluations

L07 · L08

## Store less

**Quantization**

8 or 4 bits per number

L06 (TA talk)

## Remember smartly

**Chunks + KV cache**

Look backwards only; don't recompute the past

L07 · L08

**40 ms**

per frame at 25 fps

**~200 ms**

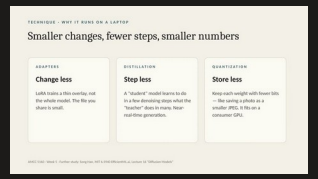
human turn gap

**0.07 → 100+**

fps: NeRF → splats

REVISIT

Click to open




L05 · 67

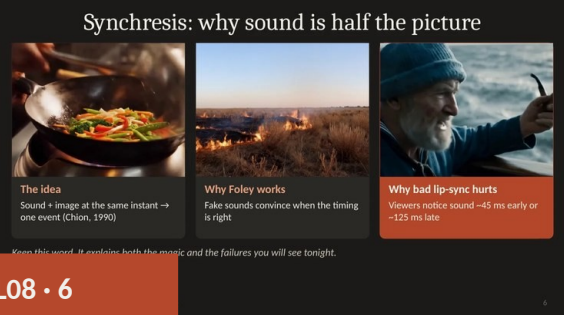



L07 · 15




L08 · 65

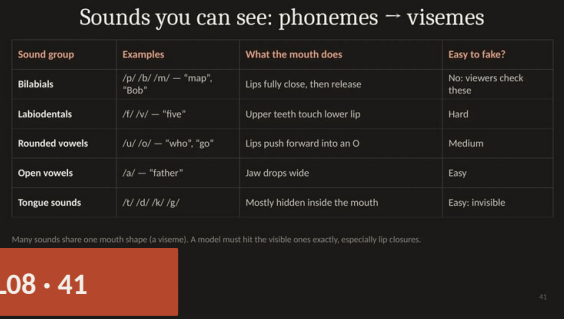
# Sound shares the clock




L08 · 6




L08 · 19




L08 · 41

1 **Synchresis (Chion, 1990)**

Sound and image at the same instant weld into one event. Off-sync is felt by everyone. (L08 · 6)

2 **Two streams, one clock**

LTX-2: 14 B video + 5 B audio; cross-attention on time positions only. (L08 · 18–21)

3 **Check p, b and m**

Bilabials need closed lips. The fastest lip-sync test there is. (L08 · 41)

# Generate the small thing, render the big thing

## VASA-1

Lecture 08

Generate a tiny motion code for pose and expression; render the face from one photo.

**40 fps**

## Genie latent actions

Lecture 09

Squeeze the change between two frames into one of 8 codes. Nobody labels anything.

**8 codes → left, right, jump**

## Gaussian splats

Lecture 09

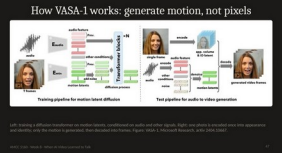
Put soft coloured blobs only where something exists; skip the empty air.

**0.07 → 100+ fps**

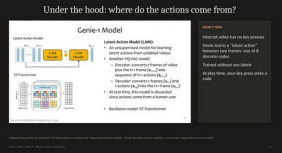
*When your project is too slow or unstable, ask: what is the small thing?*

**REVISIT**

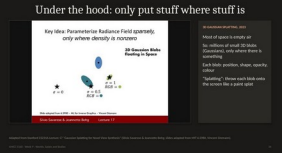
Click to open




L08 · 47

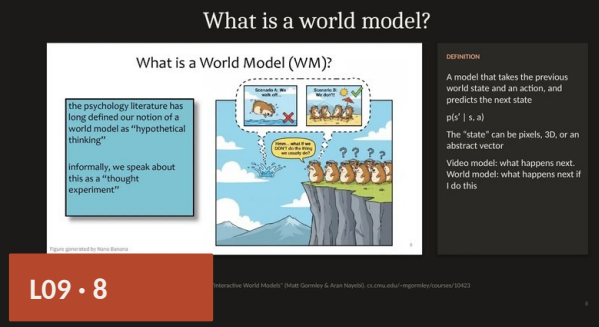



L09 · 16

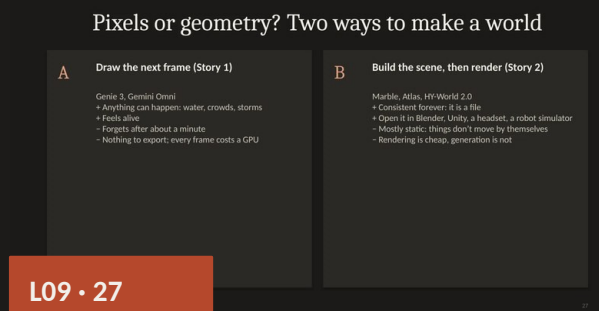



L09 · 34

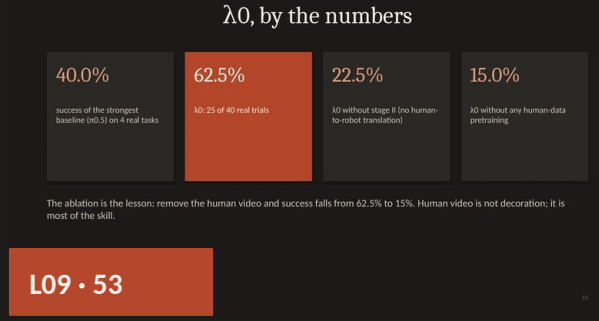
# Add an action, and video becomes a place




L09 · 8




L09 · 27




L09 · 53

1 **State + action → next state**

A video model asks “what next?”; a world model asks “what next if I do this?” (L09 · 8, 15)

2 **Pixels or geometry**

Genie draws and forgets; Marble and Atlas give you a file to open and edit. (L09 · 27)

3 **Robots learn from people**

λ0: 62.5% vs 40% baseline. Copying plateaus near 80%; practice gets past it. (L09 · 53, 59)

# Tools vanish. Keep your sources.

```
my-project/
  README.md      model · version · date · settings · one-line intent
  prompts/       every prompt kept, word for word, with its seed
  refs/          images · clips · songs, with credits and licences
  raw/           unedited generations, interesting failures too
  edit/          the edit project and the exported master
```

**Dec 2024**

Sora becomes a product

**24 Sep 2026**

API switched off, no replacement

**~18 months**

as a product

REVISIT

Click to open




L07 · 59




L07 · 65

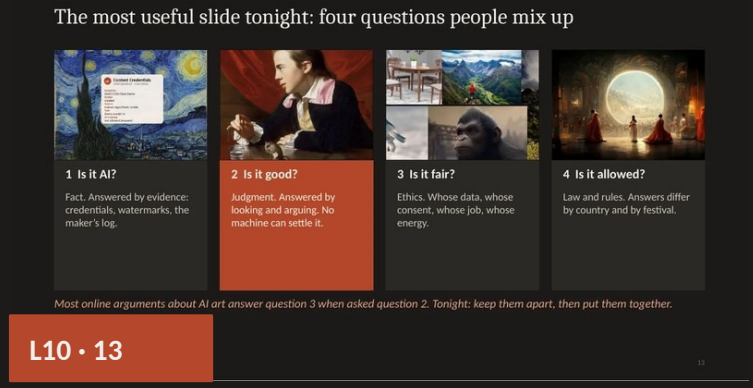


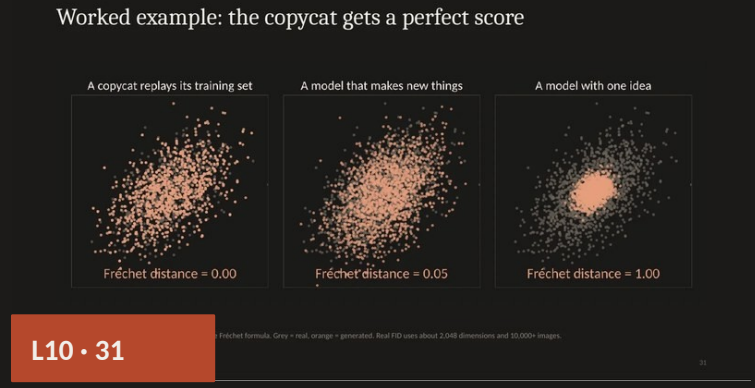

L07 · 66

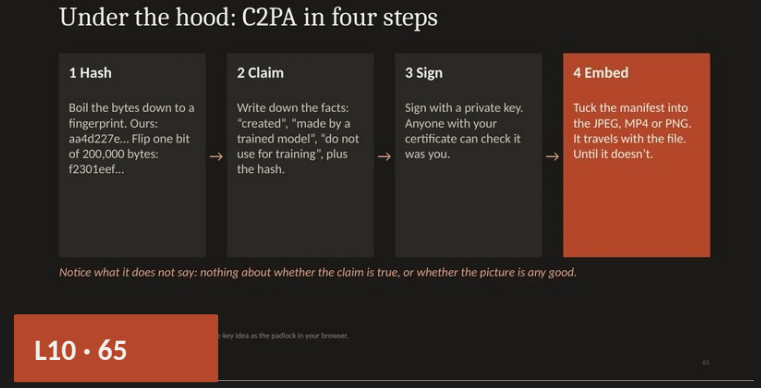
THREAD 4

# 4 Judgment: looking, measuring, signing

*Lectures 02, 04, 05, 08 and 10*



# Four questions people mix up

1

## Is it AI?

Fact · evidence

Credentials, watermarks, logs

L10

2

## Is it good?

Judgment · looking

Critique; scores are narrow

L02 · L04 · L10

3

## Is it fair?

Ethics · consent

Data, likeness, places, jobs

L05 · L08 · L09

4

## Is it allowed?

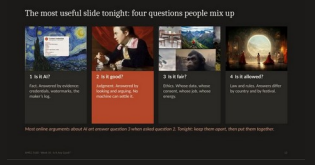
Law · rules

Lawsuits, SAG-AFTRA, EU Art. 50

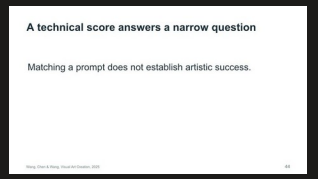
L05 · L08 · L10

REVISIT

Click to open




L10 · 13




L04 · 44

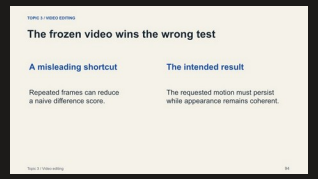
# Every score we met, and its trap

| Score | What it sees | Its trap |
|---|---|---|
| **Frame difference · L02** | Change between frames | The frozen video wins |
| **PSNR · SSIM · L10** | Pixel-by-pixel match | An 8-pixel shift scores worse than a wrecked JPEG |
| **FID · L10** | Distance between two crowds | A copycat scores a perfect 0.00 |
| **Sync-C / Sync-D · L08** | Lip-audio offset | Wav2Lip “beats” real video: trained on the judge |
| **Aesthetic score · L10** | Predicted liking | Vera Molnár falls below the cut-off |
| **Elo · L10** | Blind A/B votes | A 43-point gap ≈ 56%: nearly a coin flip |

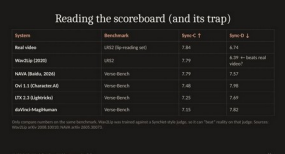
*Goodhart's law: when a measure becomes a target, it stops being a good measure.*

REVISIT

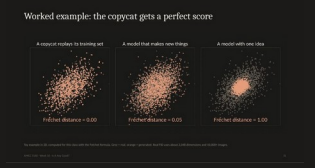
Click to open




L02 · 94




L08 · 44




L10 · 31

# Look before you judge

## See · Think · Wonder

Harvard Project Zero

1. What do you see? Facts only
2. What do you think about it?
3. What does it make you wonder?

## Feldman's four steps

Classic art criticism

1. Describe
2. Analyze
3. Interpret
4. Judge, at the very end

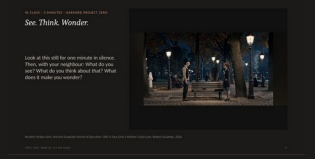
## Reading an AI image

Salvaggio, Reading 1 · L10

1. What is here too often?
2. What is missing?
3. Where does it break?
4. **What did the artist fight?**

REVISIT

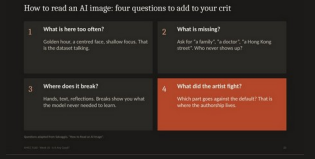
Click to open




L10 · 17




L10 · 18



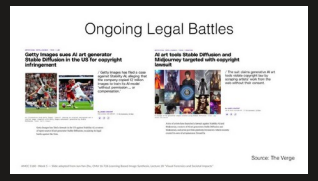

L10 · 20

# Consent has layers

| | | |
|---|---|---|
| Training data | L05 · L10 | Billions of images behind the model. Nobody in one museum can ask those artists. |
| Likeness | L08 | Voice and face. SAG-AFTRA 2026: informed consent and fair pay for replicas. |
| Places and bodies | L09 | Who owns a scan of a place? Whose homes become robot training data? |
| The work itself | L10 | Rights to the source material on screen: the layer Madison could answer. |

REVISIT

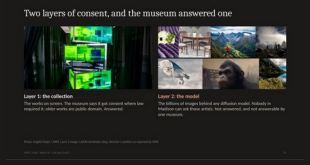
Click to open




L05 · 16




L08 · 88




L10 · 53

# Sign your work, and don't apologise

1 **Visible label**

Promise: someone told you

2 **C2PA · Content Credentials**

Hash → claim → sign → embed
Tamper-evident, but stripped by a re-save. Absence proves nothing.

3 **Pixel watermark**

A hidden pattern, e.g. SynthID

**CREDIT BLOCK · A2, A3, FINAL**

- Title · length · medium
- Director: you
- Models + versions (pinned)
- Human work, counted: 41 attempts, 6 selected, edit, grade
- References: yours, or credited
- Consent: say it, even when “no real person”
- One plain sentence of disclosure

**REVISIT**

Click to open




L10 · 64




L10 · 67




L10 · 73

# Four debates. Has your vote moved?

**L07**

If one generation is a whole 30-second scene, what is the director's job now?

**L08**

An AI performer that never existed should be allowed to win an acting award.

**L09**

A video generator is a world model.

**L10**

You run the Madison museum. Do you show Diffuse Control?

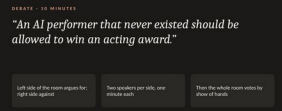
**Write: Then · Now · One piece of evidence from a different lecture**

REVISIT

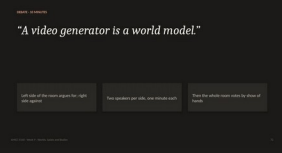
Click to open



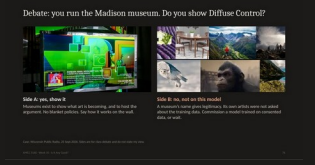

L07 · 27




L08 · 94




L09 · 72




L10 · 76

# Game: which week said it?

1 “Change, keep, adapt.” L __

2 “A seed is an address, not an instruction.” L __

3 “Frame time is not denoising time.” L __

4 “Whose model is it?” L __

5 “Plan the song, then the steps.” L __

6 “Watch the p, b and m sounds.” L __

7 “Walk away, and come back.” L __

8 “Its absence proves nothing.” L __

*90 seconds · no slides · write the lecture number*

# Answers, and where to revisit

1

## L02 · 9

The edit contract

2

## L03 · 48

Seeds

3

## L04 · 9

Two kinds of time

4

## L05 · 3

Whose model is it?

5

## L07 · 34

Wan-Dancer's two-level plan

6

## L08 · 41

Phonemes → visemes

7

## L09 · 19

The round-trip test

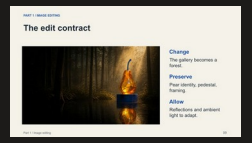
8

## L10 · 67

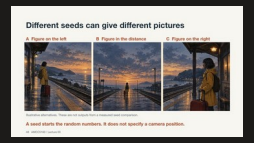
Content Credentials

REVISIT

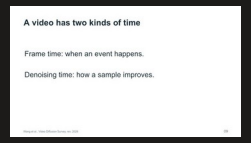
Click to open



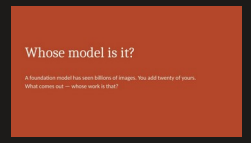
L02 · 9



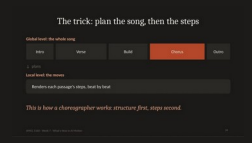
L03 · 48



L04 · 9



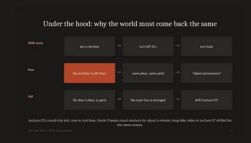
L05 · 3



L07 · 34



L08 · 41



L09 · 19



L10 · 67

# Bring the semester into your final presentation

### Explain a pipeline
Thread 1

Your own four-stage diagram, with real model names and versions

### Direct a coherent work
Thread 2

Decisions, not outputs: your edit contract or beat sheet

### Document an experiment
L01 · L03

One variable at a time; attempts counted; side by side

### Critique with evidence
Thread 4

Four steps on your own work; what did you fight?

### Account for AI use
L07 · L10

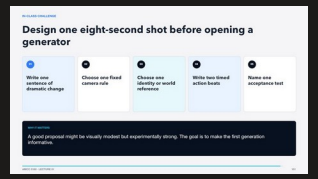
Credit block, pinned versions, consent line

### Present a refined project
Final

One clear intention, and what you would do next

**REVISIT**

Click to open




L01 · 101




L10 · 81




L10 · 73

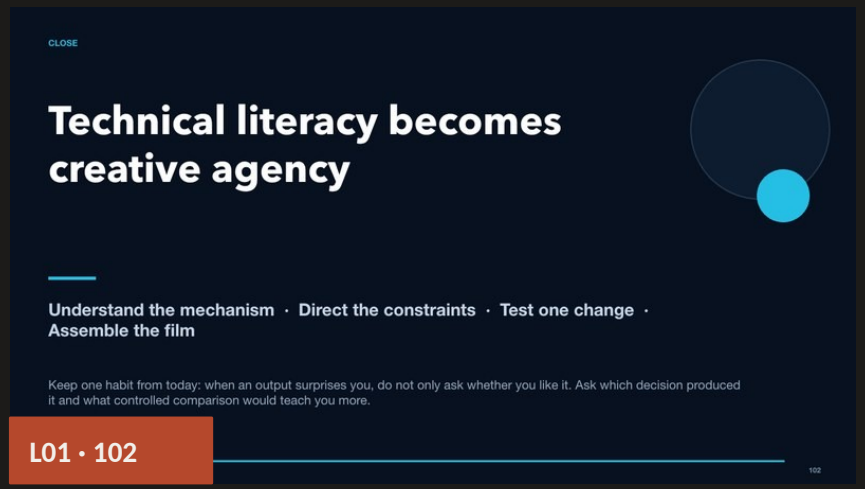
# Three weeks left



| Nov 10 | Nov 17 | Nov 24 | Nov 30 | Semester end |
|---|---|---|---|---|
| **Tonight** | **Finals I** | **Finals II** | **A2 + A3** | **Quizzes** |
| Last weekly presentations, after the break | Final presentations, sign-up page | Final presentations, sign-up page | 23:59 HKT: both assignments due | Quizzes 01–11, editable until then |

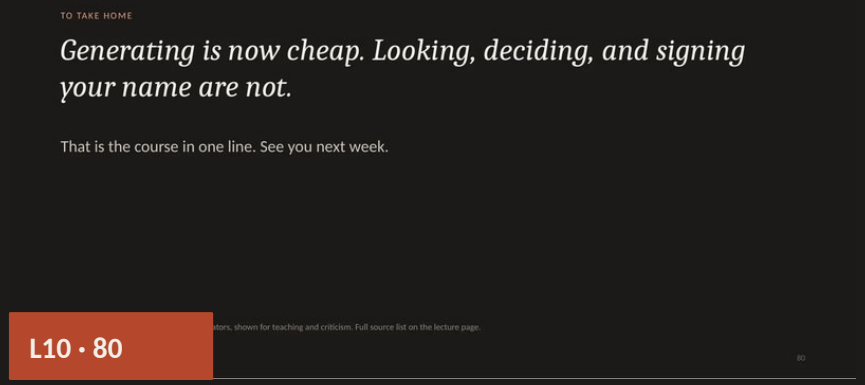
**This week**

Check your final slot and add your slides link · Revisit any game answer you missed · Start Quiz 11 (review, connects two lectures per question)




LECTURE 01

When an output surprises you, ask which decision produced it, and what controlled comparison would teach you more.




LECTURE 10

*Generating is now cheap. Looking, deciding, and signing your name are not.*

Thank you. Ten-minute break, then the last weekly presentations.